MD ASHIKUL HAQUE | CURRICULUM VITAE

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RESEARCH INTERESTS

- **Internet of Things (IoT) and Wireless Networked Systems:** anti-jamming, coexistence, and satellite connectivity for low-power wide-area networks (LPWANs).
- **Real-Time and Cyber-Physical Systems (CPS):** real-time scheduling for physical artificial intelligence (AI), from large language model (LLM) agents to connected vehicles.
- **AI for IoT and CPS:** reinforcement learning and generative AI for networked systems.

RESEARCH HIGHLIGHTS

Nine peer-reviewed conference papers, eight as first author, at venues including ACM **SenSys**, IEEE **RTSS**, IEEE **INFOCOM**, and ACM **MobiHoc**, seven of them at conferences with acceptance rates from 14% to 25%. Published the first anti-jamming work for LPWANs and built gateway-side decoders that raise packet reception under jamming by up to 106.56× with no change to the LoRa end devices. Published the first work that enables communication from Long Range Frequency Hopping Spread Spectrum (LR-FHSS) to LoRa, which improves reliability at long range by up to 79%. Designed deadline-aware token scheduling for LLM agents that control physical devices, which cut the share of critical requests that were rejected or late from 92.5% to 23.8% in a vLLM stress test.

EDUCATION

University of Texas at Dallas — **Richardson, Texas, USA**
PhD in Computer Science — *Expected May 2027*
Dissertation: *Dependable Low-Power Wide-Area Networking under Contention: Jamming, Coexistence, and Deadlines*
Advisor: **Prof. Abusayeed Saifullah** (NSF CAREER Award 2019)

Wayne State University — **Detroit, Michigan, USA**
MSc in Computer Science — *2024*

Bangladesh University of Engineering and Technology (BUET) — **Dhaka, Bangladesh**
BSc in Computer Science and Engineering — *2018*

CONFERENCE PUBLICATIONS

Note: Each published paper's title links to its full text.

[C9] "Deadline-Aware Token Scheduling for Physical-I/O LLM Agents"
Md Ashikul Haque and Abusayeed Saifullah
IEEE Real-Time Systems Symposium ***(RTSS)***, 2026
(50 papers accepted out of 350, acceptance ratio: 14.3%)
Artifact passed the RTSS 2026 Artifact Evaluation.

[C8] "Enabling Cross Technology Communication from LR-FHSS to LoRa"
Md Ashikul Haque, Aakriti Jain, Venkata Prashant Modekurthy, and Abusayeed Saifullah
ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems ***(SenSys)***,
CPS-IoT Week 2026
(52 papers accepted out of 263 in the November round, acceptance ratio: 19.8%)
To our knowledge, the first cross-technology communication from LR-FHSS to LoRa.

[C7] "Decoding LoRa Packets under Collaborative Jamming Attacks"
Md Ashikul Haque and Abusayeed Saifullah
International Conference on Embedded Wireless Systems and Networks ***(EWSN)***, 2026
(15 papers accepted out of 71, acceptance ratio: 21.1%)

[C6] "ReMix: Resource-Reclaiming Mixed-Criticality Scheduling for Industrial IoT"
Md Ashikul Haque and Abusayeed Saifullah
*International Conference on Computer Communications and Networks **(ICCCN)***, 2026
(acceptance ratio: 25%)

[C5] "Mitigating Jamming Attacks in LoRa Networks: A Defense Strategy against LoRa-Based Jammers"
Md Ashikul Haque and Abusayeed Saifullah
*ACM International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing **(MobiHoc)***, 2025
(39 papers accepted out of 169, acceptance ratio: 23.1%)

[C4] "Deep Reinforcement Learning Based Coexistence Management in LPWAN"
Md Ashikul Haque, Abusayeed Saifullah, and Haibo Zhang
*IEEE International Conference on Computer Communications **(INFOCOM)***, 2025
(272 papers accepted out of 1458, acceptance ratio: 18.7%)

[C3] "Handling Jamming Attacks in a LoRa Network"
Md Ashikul Haque and Abusayeed Saifullah
*ACM/IEEE Conference on Internet of Things Design and Implementation **(IoTDI)***, CPS-IoT Week 2024

[C2] "Burst-MAC: A MAC Protocol for Handling Burst Traffic in LoRa Network"
Aakriti Jain, **Md Ashikul Haque**, Abusayeed Saifullah, and Haibo Zhang
*IEEE Real-Time Systems Symposium **(RTSS)***, 2024
(32 papers accepted out of 140, acceptance ratio: 22.9%)

[C1] "A Game-Theoretic Approach for Mitigating Jamming Attacks in LPWAN"
Md Ashikul Haque and Abusayeed Saifullah
*International Conference on Embedded Wireless Systems and Networks **(EWSN)***, 2023
To our knowledge, the first anti-jamming work in LPWAN.

WORKSHOP PUBLICATIONS

[W2] "Coordinating Coexisting Learning Agents in Shared Spectrum via Parameter Space Complementarity"
Md Ashikul Haque, Abusayeed Saifullah, and Haibo Zhang
ICLR 2026 Workshop on Agents in the Wild: Safety, Security, and Beyond, 2026

[W1] "Scalable Real-Time Control in Industrial Cyber-Physical Systems"
Md Ashikul Haque, Dali Ismail, and Abusayeed Saifullah
Workshop on Software-Defined Networking for Cyber-Physical System Automation (SDN-CPS), co-located with ICDCN, pp. 353–358, 2024

RESEARCH FUNDING AND PROPOSALS

• Contributed to the writing of an NSF proposal on real-time cooperative perception for connected vehicles, with **Prof. Abusayeed Saifullah** as principal investigator *2026*

• My doctoral research has been supported by five NSF awards (CNS-2301757, CAREER-2306486, CNS-2306745, CNS-2601685, and CNS-2602744) and one ONR award (N00014-23-1-2151), all with **Prof. Abusayeed Saifullah** as principal investigator *2022 – Present*

AWARDS AND HONORS

• Dean's List Award, Wayne State University *2024*
• Graduate Student Professional Travel Award, College of Engineering, Wayne State University *2024*
• ACM SIGBED Travel Grant, CPS-IoT Week, San Antonio, Texas, USA *2023*

CONFERENCE PRESENTATIONS

• ReMix: Resource-Reclaiming Mixed-Criticality Scheduling for Industrial IoT
Conference talk at IEEE ICCCN 2026, Honolulu, Hawaii, USA, July 2026

• Mitigating Jamming Attacks in LoRa Networks: A Defense Strategy against LoRa-Based Jammers
Conference talk at ACM MobiHoc 2025, Rice University, Houston, Texas, USA, October 2025

• Handling Jamming Attacks in a LoRa Network
Conference talk at ACM/IEEE IoTDI 2024, CPS-IoT Week 2024, Hong Kong, China, May 2024

RESEARCH AND PROFESSIONAL EXPERIENCE

University of Texas at Dallas **Richardson, Texas, USA**
Graduate Research Assistant, Department of Computer Science *Fall 2025 and Summer 2026*
Wayne State University **Detroit, Michigan, USA**
Graduate Research Assistant, Department of Computer Science *Fall 2022 – Summer 2025*
Advisor at both universities: **Prof. Abusayeed Saifullah**
• Designed gateway-side and network-server-side defenses that recover LoRa packets under single, LoRa-based, collaborative, and spatially distributed jammers, evaluated on ten-node outdoor testbeds of USRP B200 gateways and commodity LoRa nodes with up to three USRP jammers.
• Built learning-based coexistence management at the LoRa Network Server and disclosure-free coordination among LoRaWAN operators, evaluated on an indoor USRP and Dragino testbed and in PyTorch, NS-3, and a demodulator-level simulation of 32,000 devices across 8 operators.
• Designed real-time medium access and mixed-criticality scheduling for LoRa and industrial wireless sensor-actuator networks, evaluated on LoRa devices and a 15-node Tmote Sky testbed, and built deadline-aware schedulers for LLM serving on vLLM/ROCm and llama.cpp/Vulkan and for cooperative perception on three MentorPi robot cars and in 1,020 CARLA runs.
• Built cross-technology communication from LR-FHSS to LoRa on GNU Radio and USRP B200, evaluated outdoors in a suburban area.
• Designed per-message delivery certificates for direct-to-satellite LR-FHSS, evaluated on a testbed of ten commercial LR-FHSS nodes and six USRP radios driving a calibrated emulator, and signature-bound frame recovery, evaluated in simulation and on an indoor four-USRP testbed.

Dhaka Electric Supply Company Limited **Dhaka, Bangladesh**
Assistant Engineer, ICT Division *2020 – 2021*
• Maintained the ICT infrastructure of a metropolitan electricity utility, with monitoring, incident response, and reliability-focused operation of distributed systems.

Samsung R&D Institute Bangladesh **Dhaka, Bangladesh**
Software Engineer, IoT Division *2019 – 2020*
• Developed user interface and cloud integration features of the SmartThings iOS application, which Samsung reported as serving more than 45 million monthly active users and 5,000 device types in October 2019.
• Developed production C++ components of the Samsung Cloud Client (SCClient), the shared library for cloud connectivity and device communication in SmartThings, exposed them to iOS through a Swift wrapper, and automated the build, packaging, and release workflow of the SCClient iOS team in Bash.

TEACHING EXPERIENCE

University of Texas at Dallas **Richardson, Texas, USA**
Graduate Teaching Assistant, Department of Computer Science *Spring and Fall 2026*
Courses: CS 5390 Computer Networks; CS 6352 Performance of Computer Systems and Networks
• Graded assignments, examinations, and project deliverables; held office hours; supported course projects by clarifying requirements and helping students debug implementations.

Guest Lecturer, Department of Computer Science *2025*
Course: Internet of Things (graduate)
• Delivered multiple lectures on LoRa networking and network simulation in NS-3.

Millennium University **Dhaka, Bangladesh**
Lecturer, Department of Computer Science *2018 – 2019*
Courses: Structured Programming; Data Communication
• Instructor of record for both courses: designed lectures, laboratory assignments, and examinations.

MENTORING

• Nasif Ahmed, PhD student, University of Texas at Dallas. Junior doctoral student in our laboratory. I mentor him on identifying new problems and approaching their solutions, on experiments with USRP radios and commercial off-the-shelf (COTS) LoRa devices, and on writing systems papers *January 2025 – Present*

PROFESSIONAL SERVICE

Journal Reviewer:

• ACM Transactions on Cyber-Physical Systems *2026*
• Pervasive and Mobile Computing (Elsevier) *2023*

Conference Sub-reviewer:

• IEEE International Conference on Distributed Computing in Smart Systems and the IoT (DCOSS-IoT) *2026*
• IEEE International Conference on Embedded and Real-Time Computing Systems and Applications (RTCSA) *2024, 2025*
• IEEE International Conference on Sensing, Communication, and Networking (SECON) *2026*

Conference Referee:

• ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems (SenSys) *2022, 2024, 2025, 2026*
• IEEE Real-Time Systems Symposium (RTSS) *2023, 2024, 2025, 2026*
• IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS) *2023, 2024*
• International Conference on Embedded Wireless Systems and Networks (EWSN) *2023, 2024, 2025*
• IEEE International Conference on Distributed Computing in Smart Systems and the IoT (DCOSS-IoT) *2023, 2024, 2025*

Journal Referee:

• IEEE/ACM Transactions on Networking *2024, 2025*
• ACM Transactions on Cyber-Physical Systems *2025*

Other Service:

• Student Volunteer, CPS-IoT Week 2023, San Antonio, Texas, USA *2023*

REFERENCES

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